

AIN4312
SPECIAL TOPICS IN
ARTIFICIAL INTELLIGENCE ENGINEERING II

**Controlled AI-Driven Energy Optimization Management System for Smart
Buildings**

ABSTRACT

This is a web-based AI project built to analyze and optimize energy consumption in smart buildings. The system controls a home's electronic devices such as ACs heaters and lights, switching them on and off or adjusting their operating modes to maintain optimal energy consumption without sacrificing comfort. A multi-output gradient boosting model is used to predict indoor temperature and energy consumption, these predictions are then fed into a reinforcement learning model. The RL model uses these outputs in addition to other environmental and user defined variables, to achieve the most satisfactory comfort/energy efficiency balance for the client. The results are later visualized through an interactive website, where the user can observe his/her energy savings, manage multiple homes and devices, and control the system permissions, such as excluding specific devices from automated control. This combination of predictive modeling, reinforcement learning, and user interactive website leads to environmental sustainability and cost efficiency in modern smart buildings.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form	Meaning
kWh	Kilowatt-hour	Unit of energy: 1 kW used for 1 hour = 1 kWh
kW	Kilowatt	Unit of power: rate of energy use or generation
Wh	Watt-hour	Smaller unit of energy (1 kWh = 1,000 Wh)
NNs	Neural Networks	Computational models inspired by the brain, used for pattern recognition and learning from data
RL	Reinforcement Learning	A learning paradigm where agents learn by interacting with an environment and receiving rewards
DQN	Deep Q-Network	A type of deep reinforcement learning algorithm that uses neural networks to approximate Q-values for decision-making
UI	User Interface	The visual and interactive elements users engage with (buttons, menus, screens)

1. OVERVIEW

The system aims to build user friendly website that allows its users to easily analyze the system's performance and manage their homes and devices through operations such as add, delete, rename, set permissions and many more. Each home will have a separate list of devices and permissions to guide our agent. The RL agent will be trained on each home separately to ensure optimization tailored to the specific characteristics and requirements of that environment. The agent will be trained first on simulations made by our prediction model to mitigate the potential risks associated with deploying an untrained RL agent.

TEAM

Our team consists of five AI department students each one will contribute evenly to ensure high integration and fair work experience for all.

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Table 1. List of Team Members.

1.1. Identification of the need

Traditional smart home systems often rely on fixed schedules or manual user input, which can lead to inefficient energy consumption and unnecessary costs. These inefficiencies lead to environmental harm and financial burdens, especially in today's economical environment. The proposed system utilizes AI models and agents to monitor energy usage patterns in real time, dynamically adjusting devices to minimize energy waste and reduce costs. By integrating reinforcement learning, the system continuously learns the right decisions to optimize energy consumption while maintaining occupant comfort.

1.2. Definition of the problem

The challenge lies in maintaining an optimal indoor environment while minimizing energy waste. Traditional systems depend heavily on manual adjustments which are more prone to errors because of human clumsiness and the lack of predictive intelligence to adapt to changing conditions. To address this, the proposed system employs a gradient boosting model to forecast short-term temperature and energy consumption trends, providing the reinforcement learning (RL) agent with predictive insights. The RL model then uses this information to autonomously manage device operations ensuring occupant comfort while optimizing energy consumption dynamically and efficiently.

1.2.1. Functional requirements

List of the system's main functions:

- **Devices management:** This will handle all devices' action (add devices, remove devices, assign permission, set the maximum kWh consumption).
- **Home management:** This will handle all possible home actions e.g. (add home, delete home, rename home).
- **Home reading:** This function collects data from sensors for future predictions.
- **Home simulator:** this function will mimic a home environment if the available data is not enough.
- **RL agent:** This function will be responsible for creating the DQN and initializing the weights and setting the parameters for convergence.
- **Web Interface:** A user friendly website that visualizes the performance and manages the system.

1.2.2. Performance requirements

The prediction of the system should follow a specific policy, which will directly affect the total rewards returned to the RL agent. We will tackle the policy function to ensure that RL model never takes actions that increase the reward total in favor of overall comfort rate in the house.

- The prediction model should achieve a high level of accuracy (0.8 or higher).
- The RL system should achieve the best comfort / efficiency balance.
- Model response time must remain under one second to ensure real-time adaptability.
- The website should be accessible and user friendly.

1.3 Conceptual Solution

The system combines a predictive layer (based on XGBoost regression) that estimates hourly energy usage and temperature, and a decision layer (based on Deep Q-Network reinforcement learning) that chooses the optimal device actions. The learning loop continuously improves decision accuracy based on the reward/penalty policy. To display the results of our system we need to build a web based central management system with homes and devices for our models to optimize and a dashboard to display and visualize the results.

1.3.1 Literature Review

Studies in intelligent energy management highlight reinforcement learning as an effective optimization approach. Research published in IEEE (2023) demonstrated that DQN based controllers reduced HVAC energy consumption by up to 25%. Similarly, Elsevier (2024) reported that AI-driven predictive systems maintained stable indoor comfort while cutting costs significantly. MIT and Google DeepMind experiments on adaptive energy control validated the use of RL for minimizing peak loads while ensuring system stability. These findings support the integration of XGBoost for forecasting and reinforcement learning for dynamic control in this project.

1.4 Architecture

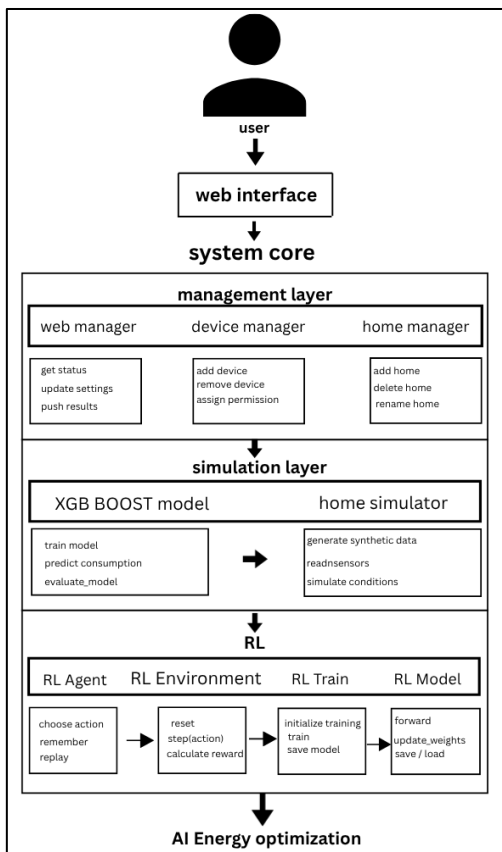


Figure 1. Imagine system Architecture.

1. Homes and Devices management – impact calculation: Manages metadata and real-time status of homes, appliances, and sensors.
2. Prediction layer: Forecasts energy demand, device usage, or environmental variables.
3. RL initializing and training: Initializes reinforcement learning agents with environment parameters.
4. Logging the training outputs: Captures metrics, rewards, policy evolution, and agent decisions.
5. RL testing layer – simulator: Evaluates trained agents in virtual environments.
6. Web visualizing: Enables user-friendly exploration of predictions, RL behavior, and device impact.

2. WORK PLAN

2.1. Work Breakdown Structure (WBS)

The project was divided into 6 main stages, and it is illustrated below.

Phase No.	Duration in weeks	Key Activities
1	1-3	Structuring, Data collection
2	4-5	Data cleaning, Prediction model training
3	6-7	Home & Devices Mang., log system
4	8-10	RL agent, RL training, RL testing
5	11-12	APIS, code cleaning
6	13-14	Web Manag. site

Table 2. Phases Structure.

2.2. Team Responsibilities

Phase No.	Responsibilities
1	AWS: Project Structuring. ALHASSAN: Project Structuring. ABDULRAHMAN: Related Work searching. HASHEM: Data Collection. SULAIMAN: Data Collection.
2	AWS, ALHASSAN, ABDULRAHMAN: Same phase 1 with documentation. HASHEM: Model Training. SULAIMAN: Data Cleaning & Merging.
3	AWS: Building Device Manag. ALHASSAN: Building Homes Manag. ABDULRAHMAN: Organize the files and fixing debugs and implement structure. HASHEM: Test the functions integrations. SULAIMAN: Implement the log files for next training.
4	AWS: Test the home environment simulator. ALHASSAN: Search for simple RL agents. ABDULRAHMAN: Build the RL agent. HASHEM: Build the train loop and save the result using log structure. SULAIMAN: Build a simulator to test the RL performance.
5	AWS: Code cleaning. ALHASSAN, HASHEM: Documenting the previous works ABDULRAHMAN, SULAIMAN: Build the APIs & testing.
6	ALHASSAN: Building the Web. ABDULRAHMAN: API & Web integration. HASHEM: Testing the web edge cases. AWS, SULAIMAN: Documenting.

Table 3. Work Distribution.

APPENDIX A

We started with the predictive model by collecting, cleaning, and merging 3 datasets. We dropped empty rows and forward filled some empty values and formulated the datasets into hourly format and matched the timestamps for accurate merging.

Timestamp	Temp (C)	Humidity (%)	CO (ppm)	HVAC Temp (C)	Energy (kWh)
2024-02-05 08:00	21.87	37.7	407	21.4	1.32
2024-02-05 08:15	24.75	43.1	430	24.7	0.90
2024-02-05 08:30	23.65	48.1	426	24.2	0.72
2024-02-05 08:45	22.99	45.9	365	23.6	1.01
2024-02-05 09:00	20.78	47.0	364	20.3	1.04

Table 4. Indoor Environment Dataset (Kaggle Source).

Explanation: This dataset includes records of temperature, humidity, and CO concentration readings, essential for simulating internal building behavior.

Time	Temperature (C)	Humidity (%)	Wind Speed (m/s)
2024-01-01T00:00	9.9	82	9.2
2024-01-01T01:00	8.7	90	6.1
2024-01-01T02:00	8.4	89	5.9
2024-01-01T03:00	8.6	87	7.2
2024-01-01T04:00	8.9	86	7.7

Table 5. Outdoor Weather Dataset (Open-Meteo API).

Explanation: The external environment dataset tracks weather patterns that influence indoor temperature and system performance.

Datetime	Global Active Power (kW)	Voltage (V)	Global Intensity (A)	Sub 1	Sub 2
2006-12-16 17:24:00	4.216	234.8	18.4	0	1
2006-12-16 17:25:00	5.360	233.6	23.0	0	1
2006-12-16 17:26:00	5.374	233.2	23.0	0	2
2006-12-16 17:27:00	5.388	233.7	23.0	0	1
2006-12-16 17:28:00	3.666	235.6	15.8	0	1

Table 6. UCI Household Energy Consumption Dataset.

Explanation: Provides detailed measurements of household power usage and serves as the baseline for performance comparison.

We tried different approaches for predicting indoor temp and energy consumption, such as LSTMs and random forest and found that gradient boosting gave us the best results (shown below).

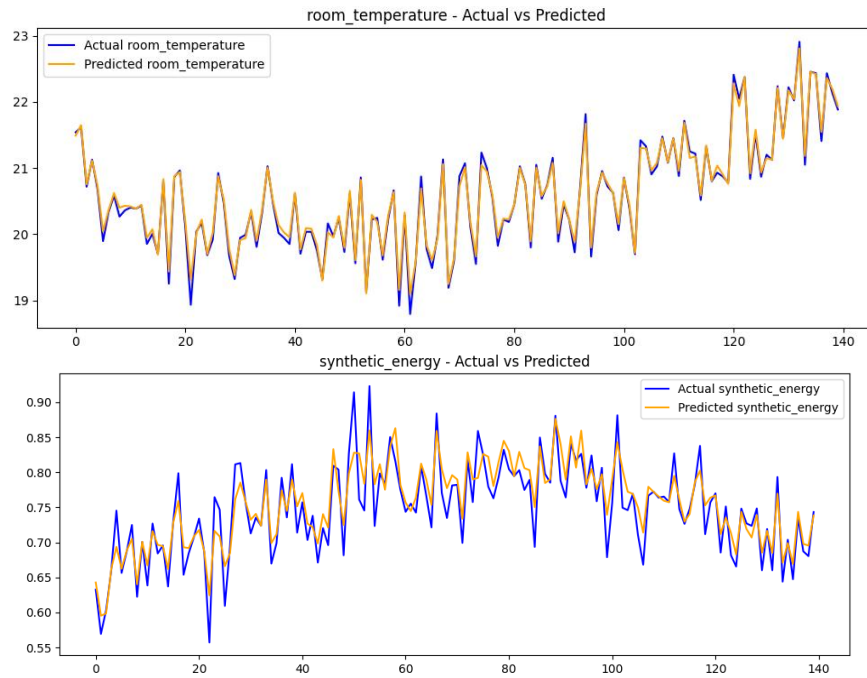


Figure 2. Model Prediction Accuracy (Actual vs Predicted).

Explanation: The XGBoost multi-output regression achieved near-perfect prediction for temperature and high precision for energy consumption, confirming its suitability for real-time forecasting.

Future Work

Next, we are building the RL agent and function using the previous model's outputs in addition to other environmental and user defined data while other team members design the website.